

Effect of Ionic Liquid Confinement on Gas Separation Characteristics

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S Supporting Information

ABSTRACT: Work in our laboratory has focused for a number of years on examining the potential of room-temperature ionic liquids for post-combustion carbon capture processes. Results from studies of carbon dioxide solubility, diffusivity, and permeation across supported ionic liquid membranes have raised questions about the impact of a solid interface on the properties of ionic liquids. In this paper, we report results from measurements of carbon dioxide uptake into ionic liquids confined within a ceramic nanoporous film and compare carbon dioxide solubility and diffusivity to values measured with bulk phase ionic liquids. Results show that both solubility and diffusivity are enhanced in confined ionic liquids when compared to values observed in unconfined liquids. These observations have implications for gas separation processes involving supported ionic liquid membranes.

1. INTRODUCTION

Room-temperature ionic liquids (RTILs) are salts consisting of a bulky cation and an inorganic anion with melting points below 100 °C. The large cation size allows for delocalization and screening of charges, resulting in a reduction in the lattice energy and, thereby, the melting or glass transition temperature. In recent years, there has been increased interest in the potential of ionic liquids for a variety of applications, such as chemical synthesis, catalysis, electrochemical processes, and gas separations.^{1–5} Because ionic liquids have no measurable vapor pressure, solvent losses can be minimized when these materials are used in a separation process or in electrochemical devices. Ionic liquids can solubilize a wide range of compounds and have properties that can be tailored by appropriate choice of anion and cation. Ongoing work in our laboratory has focused on examining the potential of ionic liquids for carbon dioxide capture processes.^{6–11}

Many ionic liquids have an imidazolium-based cation with different alkyl side chains. In this paper, the acronym $C_nC_m\text{im}$ is used to represent an imidazolium ring with alkyl chains of length n and m carbons on the nitrogen atoms. The RTILs used in this work have bis(trifluoromethanesulfonyl)imide as the anion, which is given the acronym Tf_2N in this paper.

Many applications of ionic liquids involve the contact of these unique fluids with solid surfaces, often in the form of porous solid supports. For example, a number of investigators have examined supported ionic liquid phase catalyst systems where a transition-metal catalyst is dissolved in a thin ionic liquid film that is supported on a solid surface by covalent attachment or physisorption.^{12–18} Others, including our group, have examined supported ionic liquid membranes (SILMs) and polymeric ionic liquid membranes for gas separations.^{7,11,19–37}

While interest in various aspects of ionic liquid chemistry has grown significantly in the past 10 years, efforts to examine the interfacial properties of ionic liquids have arisen more recently, with sometimes contradictory observations. Using atomic force microscopy (AFM), Bovio et al.³⁸ found evidence of solid-like layering in thin films of the ionic liquid $C_4C_1\text{imTf}_2\text{N}$ on mica, amorphous silica, and oxidized Si (110) surfaces. These

organized structures were observed in films up to 50 nm thick. A number of investigators have examined confinement of ionic liquids in porous oxide networks (“ionogels”) prepared using a sol–gel process conducted in the ionic liquid phase. Bideau et al.³⁹ measured relaxation times using ^1H nuclear magnetic resonance (NMR) of ionic liquids confined in silica ionogels with pore diameters of 12 and 15 nm and found liquid-like behavior at temperatures below the bulk crystallization temperature. Néouze et al.⁴⁰ used differential scanning calorimetry (DSC) and ^1H NMR to examine the properties of ionic liquids confined in ionogels that were modified with hydrophobic methyl groups. While relaxation times indicated ionic liquid behavior intermediate between liquid and solid, measured conductivity values were characteristic of bulk liquid. The effect of the ionic liquid structure on interfacial organization on mica, graphite, silica, and gold surfaces was examined by Hayes et al.⁴¹ using AFM. It was found that ionic liquids that form organized structures in bulk solution, with polar and nonpolar domains, show strong organization near surfaces. Göbel and co-workers^{42,43} characterized functionalized silica monoliths and examined properties of ionic liquids that filled these porous structures, with pore sizes ranging from 2.5 to 30 nm. Infrared (IR) spectra, small-angle X-ray scattering (SAXS), X-ray diffraction (XRD), and NMR did not indicate significant structural difference between bulk and confined ionic liquid. However, DSC measurements indicated that the monolith surface affected the phase transition characteristics of these liquids. Kanakubo et al.⁴⁴ observed pore-size-dependent melting point depression for a number of different ionic liquids confined in controlled pore glasses with pore sizes ranging from 2 to 15 nm. The sensitivity of melting point

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depression to pore size was found to be significantly stronger than that observed for water in controlled pore glass.

In contrast to most observations for ionic liquids within porous oxide structures, Chen et al.⁴⁵ report that the ionic liquid $C_4C_1\text{imPF}_6$ undergoes a unique phase transition when confined within multiwalled carbon nanotubes, forming a polymorphous crystal with a melting point above 200 °C compared to its bulk melting temperature of 6 °C. A similar high melting phase was identified with atomistic molecular dynamics simulations for the ionic liquid $C_1C_1\text{imCl}$ when confined between parallel graphite walls.⁴⁶

Atomistic simulations of $C_6C_1\text{imTf}_2\text{N}$ confined in carbon nanotubes (CNTs) indicate highly ordered structures within the CNT, with cations segregated at the tube wall and anions at the tube center.⁴⁷ This nanoscale organization results in an increase in ionic liquid self-diffusivity and the solubility and diffusivity of both CO_2 and H_2 in the ionic liquid when compared to bulk phase properties. Confinement in the CNT also increases the CO_2/H_2 selectivity of this ionic liquid.

While a number of studies have been performed to examine SILMs for gas separations,^{7,19–37} only a few of these investigations have considered the effect of the support surface on the characteristics of the ionic liquid. Neves et al.³⁶ examined gas separation in SILMs prepared with two different PVDF support membranes, one being hydrophobic and the other being hydrophilic. Results showed that SILMs prepared with the more hydrophobic support were more stable than those prepared with the hydrophilic support, possibly because of better affinity between the RTIL and the support.

For a number of years, work in our laboratory has focused on measurements of gas solubility and diffusivity in RTILs and relating these properties to the RTIL structure.^{6,8,9} Our efforts focused primarily on carbon dioxide. The experimental system used for determining gas solubility and diffusion involved tracking the decrease in pressure that results following the introduction of target gas into a small closed chamber containing a film of ionic liquid. The solubility and diffusivity of carbon dioxide were determined by fitting the pressure decay to a one-dimensional diffusion model, with gas solubility characterized using Henry's law constant.

Work in our laboratory has also involved measurements of CO_2 and N_2 permeation through SILMs.¹¹ These SILMs were prepared by impregnating porous anodic alumina films with RTILs. Membrane permeation experiments were conducted with pure CO_2 and pure N_2 , yielding gas permeance values and ideal selectivities, with CO_2/N_2 selectivities ranging from 10 to 20. The pores in the anodic alumina support membranes are parallel capillaries with a tight pore size distribution. Therefore, the pore geometry can be easily characterized, enabling us to compare measured membrane permeance to values predicted from the solubility and diffusivity values determined from gas uptake measurements into bulk phase ionic liquid, along with the membrane thickness (60 μm). A comparison of the measured and predicted CO_2 permeance values is shown in Figure 1. For all ionic liquids, membrane permeance values were higher than values predicted from the gas uptake measurements. In addition, a comparison of CO_2 permeance values in 20 nm pore size membranes to values measured in 100 nm pore size membranes shows permeance in the larger pore membranes to be a factor of ~ 2 larger than values measured in the smaller pore membrane.¹¹ The permeance used when making this comparison is based on the ionic liquid area and not the total membrane area. Therefore, the observed

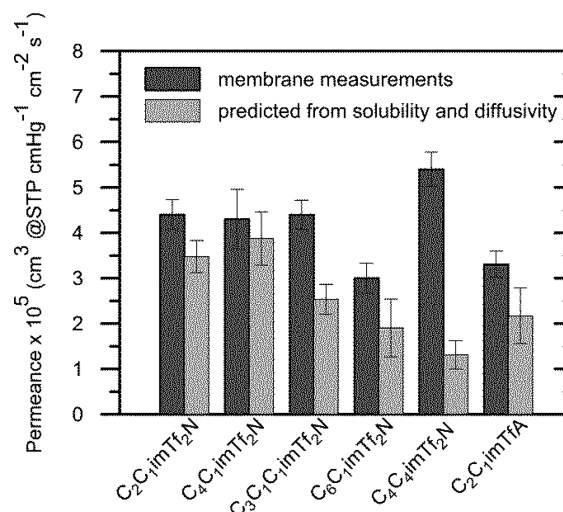


Figure 1. Comparison of CO_2 permeance measured with SILMs to values predicted from bulk phase gas uptake measurements. Membrane permeation measurements were performed with 20 nm pore size anodic alumina membranes. All measurements were performed in the Baltus lab.^{8,9,11} The anion TFA is trifluoroacetate.

differences in permeance represent differences in ionic liquid characteristics in the two membrane systems. CO_2/N_2 selectivity was found to be independent of the membrane pore size, indicating that ionic liquid property changes resulting from confinement do not appear to uniquely impact CO_2 permeability.

CO_2 permeability values measured in our laboratory with anodic alumina support membranes are consistently larger than values reported by Scovazzo and co-workers^{20,21} for SILMs prepared from porous glass fiber supports and by Zhao et al.⁴⁸ for SILMs prepared with polyethersulfone support membranes [permeability was calculated from measured permeance using the membrane thickness (60 μm)].

These results provided us with motivation to more carefully examine the properties of ionic liquids when confined within a porous support. In this paper, we report results from CO_2 uptake measurements into bulk ionic liquids and into ionic liquids confined within a porous ceramic support. CO_2 solubility, reported as the Henry's law constant, and diffusivity in confined and bulk phases are compared.

2. EXPERIMENTAL SECTION

Two different ionic liquids were studied in this work: 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ($C_2C_1\text{imTf}_2\text{N}$, Flukar, >97% purity) and 1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ($C_6C_1\text{imTf}_2\text{N}$, IOLITEC, 99% purity). Each ionic liquid sample was dried at 80 °C and 25–30 Hg vacuum for at least 8 h prior to gas uptake measurements. High-purity (99.995%) carbon dioxide was obtained from Merriam-Graves Co. (Charlestown, NH).

For gas uptake measurements into confined ionic liquid, a porous ceramic membrane [Sterlitech (P/N 47U300) with a $\text{ZrO}_2\text{--TiO}_2$ selective layer and a TiO_2 support layer] was used to confine the ionic liquid. The membrane has a molecular weight cut-off (MWCO) of 300 kD and thickness of 250 μm . The average pore size in the selective layer in these membranes can be estimated to be $\sim 20\text{--}30$ nm based on the hydrodynamic size of 300 kD dextrans.⁴⁹ The 47 mm diameter ceramic membranes were cut using a diamond core drill into smaller disks that just fit into the gas uptake sample chamber.

The experimental system used for gas uptake measurements is the same as used by our group in previous work and involves tracking the

decrease in pressure that results following the introduction of target gas (CO₂ in this work) into a small closed chamber containing RTIL. Details of the experimental setup and operating principles for our measurements are described in detail elsewhere.^{6,8}

By combining Fick's law for one-dimensional diffusion in the RTIL film with a mass balance on gas above the liquid, the pressure decay can be related to system parameters and gas properties by

$$\ln \frac{P}{P_0} = \frac{8}{\pi^2} \frac{V_{\text{IL}}}{V} \frac{\rho_{\text{IL}}}{MW_{\text{IL}}} \frac{RT}{H} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \left[\exp\left(-\frac{(2n+1)^2 \pi^2 D t}{4L^2}\right) - 1 \right] \quad (1)$$

where P_0 is the initial pressure in the sample chamber, V is the volume of gas, V_{IL} is the volume, ρ_{IL} is the density, MW_{IL} is the molecular weight, and L is the thickness of the RTIL film, and H is the Henry's law constant, and D is the diffusivity of CO₂ in the ionic liquid. In developing eq 1, it is assumed that the physical properties (i.e., density and viscosity) of the ionic liquid film do not change during the gas dissolution process. For bulk phase measurements, sample sizes of 0.25 mL of C₂C₁imTf₂N and 0.23 mL of C₆C₁imTf₂N were used, yielding film thicknesses of 245 and 225 μm for C₂C₁imTf₂N and C₆C₁imTf₂N, respectively.

Mathematical details of the derivation of eq 1 are presented in Hou and Baltus.⁶ The final model includes two unknowns, H and D . Using a nonlinear least-squares method in the Matlab curve fitting tool box, the experimental P versus t data were fit to eq 1 to determine H and D , including 55 terms in the summation.

The same experimental system was used to monitor gas uptake into confined ionic liquid with an ionic liquid imbibed ceramic membrane replacing the bulk ionic liquid sample. The ceramic film was saturated with ionic liquid by placing the film onto a layer of ionic liquid in a Petri dish, which was under vacuum in a desiccator, and allowing the ionic liquid to permeate through the film. It was noted that the upper membrane face changed from white to glossy as ionic liquid appeared at this interface.

The kinetics of imbibition were tracked by monitoring weight versus time, finding that it took ~ 10 h for the ionic liquid to saturate the membrane. Once a steady mass was achieved during the imbibition process; it is reasonable to assume that ionic liquid completely filled the pore space. The validity of this assumption was also confirmed by comparing the mass of imbibed C₆C₁imTf₂N to the mass of octanol imbibed in the same film (after ionic liquid was removed from the film). The pore volume was determined from the ratio of imbibed fluid mass to fluid density, assuming that the density of imbibed fluid is the same as the bulk value. The pore volumes occupied by these two fluids agreed within experimental error (2%). Because the surface tension of these two fluids is different, it is unlikely that these volumes would be in agreement if the fluids did not completely fill the pore space.

To monitor gas uptake into confined ionic liquid, the membrane with imbibed ionic liquid was placed in the sample chamber, and gas uptake was monitored as previously described for the bulk phase measurements. For each experiment, the mass of ionic liquid (the product $V_{\text{IL}}\rho_{\text{IL}}$ in eq 1) in the membrane was known by weighing the disk before and after impregnation with the liquid.

The path length for diffusion for the measurements with confined ionic liquid is longer than the thickness of the ceramic membrane because of the complex pore geometry in these membranes. Therefore, L in eq 1 is replaced by τL , where L is the membrane thickness and τ is the tortuosity of the porous system. An approach previously used by Morgan et al.¹⁹ was used to determine the tortuosity. Here, CO₂ uptake into confined octanol was measured using the same experimental system and approach as used for the RTIL measurements. Using literature values for the solubility ($0.059 \text{ mol L}^{-1} \text{ atm}^{-1}$) and diffusivity of CO₂ in octanol ($1.46 \times 10^{-5} \text{ cm}^2/\text{s}$),^{19,50,51} the pressure versus time data were fit to eq 1 to determine the tortuosity, yielding $\tau = 1.7$.

All gas uptake measurements were performed at 25 °C. The initial pressure in the sample chamber was ~ 2 bar, conditions where Henry's

law is expected to be valid. For each system (RTIL, bulk, or confined), 2–4 repeat experiments were performed.

3. RESULTS

Carbon dioxide solubility is reported in terms of Henry's law constant, H . Values determined in this study for the bulk ionic liquids are summarized in Table 1. Values are compared to previously reported values found in the literature. Reasonable agreement is found between values measured in this study and values reported by others.

Table 1. Henry's Law Constants for CO₂ in Bulk Phase C₂C₁imTf₂N and C₆C₁imTf₂N at 25 °C

RTIL	H (bar)		
	this work	literature value	reference
C ₂ C ₁ imTf ₂ N	39.0 $\pm$ 0.5	31.3 $\pm$ 0.4	Moganty et al. ⁸
		39.1 $\pm$ 1.0	Finotello et al. ⁶⁰
		35.6 $\pm$ 1.4	Cadena et al. ⁶¹
C ₆ C ₁ imTf ₂ N	24.6 $\pm$ 0.1	28.2 $\pm$ 0.6	Moganty et al. ⁸
		31.6 $\pm$ 0.2	Muldoon et al. ⁶²
		34.0 $\pm$ 0.3	Finotello et al. ⁶⁰

A comparison of Henry's law constants for CO₂ in bulk and confined ionic liquids is shown in Figure 2. A similar comparison for CO₂ diffusivity is shown in Figure 3. All values are listed in the Supporting Information.

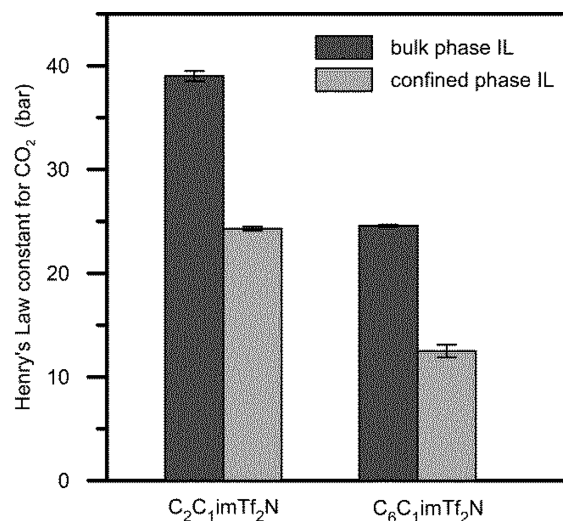


Figure 2. Comparison of Henry's law constants for CO₂ in bulk phase ionic liquid and ionic liquid confined in ceramic porous film for C₂C₁imTf₂N and C₆C₁imTf₂N. Measurements were performed at 25 °C.

The validity of eq 1 to describe gas uptake is provided by comparing measured and predicted pressure values from one experiment, as shown in Figure S1 of the Supporting Information. The excellent agreement between model and prediction over the entire experimental time supports the validity of the assumptions made in developing eq 1. Additionally, best fit values for H and D were determined by analyzing pressure data collected over different experimental time periods, 4, 8, and 10 h.⁵² Differences in fitted parameters were found to be statistically insignificant, again confirming that

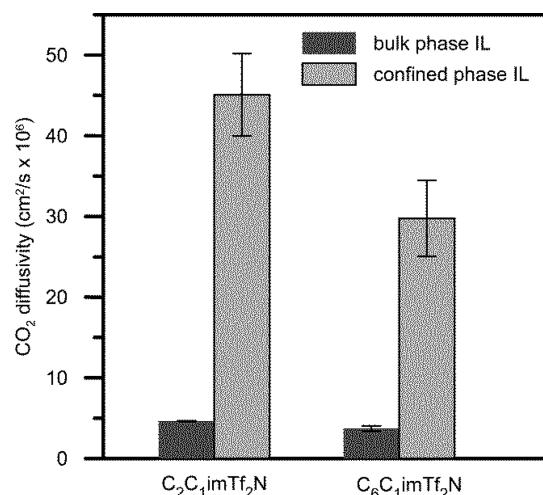


Figure 3. Comparison of CO₂ diffusivity in bulk phase ionic liquid and ionic liquid confined in ceramic porous film for C₂C₁imTf₂N and C₆C₁imTf₂N. Measurements were performed at 25 °C.

the model used to develop eq 1 is valid and that the properties of the ionic liquid do not change over the course of the measurement.

Measurements were also performed to examine CO₂ adsorption onto the ceramic film without any ionic liquid. No measurable decrease in the gas pressure was observed, indicating minimal adsorption of CO₂ to the solid surface.

4. DISCUSSION

When considering the Henry's law constant values reported here, it is important to note that increasing gas solubility corresponds to a decrease in H . A comparison of the H values listed in Table 1 and presented in Figure 2 shows higher CO₂ solubility in C₆C₁imTf₂N compared to C₂C₁imTf₂N in both bulk and confined phases. The trend of increasing gas solubility (on a mole fraction basis) with increasing alkyl chain length on the imidazolium cation has been reported previously by our group and others and has been attributed to an increase in free volume with longer alkyl chain lengths on the imidazolium cation.^{8,53,54} The results in Figure 2 show that CO₂ solubility is higher in the confined ionic liquids than in the bulk liquids, with similar ratios in the two ionic liquids (1.6 for C₂C₁imTf₂N and 1.9 for C₆C₁imTf₂N).

These observations are qualitatively consistent with results from molecular simulations reported by Shi and Sorescu, who found higher CO₂ and H₂ solubilities in C₆C₁imTf₂N confined within carbon nanotubes when compared to bulk phase solubilities.⁴⁷ Simulations enable one to examine the nanoscale structure, and Shi and Sorescu report segregation of cation and anion, with the imidazolium anions found preferentially at the tube center and the cations located near the nanotube walls. While the surface characteristics of the ceramic support used in our measurements are different from the characteristics of the CNT surface, both surfaces appear to have a similar effect on gas solubility. The tube diameter of the (20,20) CNT systems considered in these simulations is 2.7 nm, considerably smaller than the 20–30 nm pore diameter in the ceramic support membranes used in our experiments. Surface effects should be more important for smaller pore systems. However, our experimental observations indicate that the penetration depth

of cation and anion organization emanating from the pore wall appears to be significant.

The results for CO₂ diffusivity presented in Figure 3 show significantly faster gas diffusion in the confined ionic liquid compared to diffusion in bulk liquid, with again similar ratios in the two ionic liquids (9.7 for C₂C₁imTf₂N and 8.0 for C₆C₁imTf₂N). Our previous results have shown an inverse square root dependence of CO₂ diffusivity upon ionic liquid viscosity, a trend that is consistent with CO₂ diffusion in more traditional solvents.⁹ If we attribute all of the increase in diffusivity upon ionic liquid confinement to a change in viscosity, ~2 orders of magnitude decrease in viscosity is needed to explain the observed change in diffusivity.

We can again consider results obtained from molecular simulations to explain these experimental observations. Simulations of the self-diffusion coefficient of bulk C₆C₁imTf₂N and this RTIL confined in (20,20) CNT indicate that the self-diffusion coefficient of the confined RTIL is ~100 times larger than the bulk phase value.⁴⁷ Simulations of C₁C₁imCl confined between parallel walls shows the self-diffusion coefficient ~2–3 times larger when confined relative to the self-diffusivity in bulk.⁵⁵ The faster self-diffusion can be attributed to weaker ion–ion interactions upon confinement. The Stokes–Einstein equation predicts an inverse proportionality between diffusivity and viscosity; therefore, the increase in RTIL self-diffusivity with confinement is expected to be reflected in a similar decrease in viscosity and an increase in CO₂ diffusivity.

The different surface characteristics as well as the smaller size scales considered in the simulations make it difficult to quantitatively compare our observations to the simulation results. However, there is qualitative agreement between our results, which show an order of magnitude increase in CO₂ diffusivity when C₂C₁imTf₂N and C₆C₁imTf₂N were confined and the changes in RTIL self-diffusivity revealed by the molecular simulations.

The results from our previous work presented in Figure 1 show measured membrane permeance to be ~20–70% larger than the values predicted from bulk phase measurements, with the exception of C₄C₄imTf₂N, which shows a factor of 4 increase in membrane permeance relative to the prediction from bulk phase measurements. According to the solution-diffusion model, membrane permeability is the product of the gas solubility and diffusion coefficient. Therefore, our current solubility and diffusion results can be combined to estimate the impact of confinement on membrane permeability by comparing the ratio D/H (i.e., the product of diffusivity and solubility) for confined to bulk phase systems. Both ionic liquids yield the same ratio of permeability for confined IL to permeability in the bulk phase system, with over a factor of 16 increase in permeability with ionic liquid confinement (values are listed in the Supporting Information). This enhancement in permeability is considerably higher than the ratio observed in our previous SILM measurements but is qualitatively consistent with our previous results. Differences may be due to different surface characteristics in the anodic alumina membranes used as the SILM support and the ceramic membranes used in these measurements or to uncertainties in pore geometry in the anodic alumina membranes that are used when comparing measured membrane permeance to permeance predicted from bulk phase characteristics.

Using COSMOtherm to estimate free volume and a large collection of gas solubility data in ionic liquids, Shannon et al.⁵⁶ examined the relationship between gas solubility and selectivity

and ionic liquid free volume. Their analysis indicates that CO₂ solubility and CO₂ selectivity over CH₄ and N₂ increase as free volume in the ionic liquids increases and ionic liquid molar volume decreases. Applying these concepts to the results reported here indicates that free volume must be increasing upon confinement, assuming that the molar volume or density is not significantly impacted by the solid interface. A change in free volume is not unexpected if there is a reorganization of the ionic liquid structure and the relative positions of cations and anions in response to interactions with the pore surface.

Ho and co-workers examined CO₂ solubility in non-ionic physical solvents, comparing results obtained for bulk phase solvent and solvent confined within different inorganic porous supports, which they called “hybrid adsorbents”.^{57–59} Grand canonical Monte Carlo simulations were also performed on a slit pore model to understand the molecular level mechanisms responsible for observed differences between bulk and confined solvents. Results revealed enhanced CO₂ solubility for some but not all of the hybrid adsorbents, relative to solubility in both bulk solvents and “raw” solids. CO₂ solubility in a collection of hybrid adsorbents prepared with various solvents confined in porous MCM-41 was compared, with only the adsorbent prepared with the solvent *N*-methyl-2-pyrrolidone (NMP) exhibiting CO₂ solubility higher than that observed with the raw MCM-41. CO₂ solubility in a hybrid system with NMP confined in porous alumina was 5 times larger than that of the bulk solvent and 20% larger than that of raw alumina. Simulation results indicate that the enhanced solubility results from layering of CO₂ within the pores. The solvent size plays a major role, with solvent layering creating micropore channels for CO₂ adsorption. The micropore channels identified in this study may simply be an indicator of free volume resulting with confinement. A similar phenomenon may be responsible for the enhancement in CO₂ solubility and diffusivity observed in this study with confinement of the ionic liquids.

5. CONCLUSION

The results reported in this paper show that both solubility and diffusivity of CO₂ in RTILs are enhanced when the RTIL is confined within a porous support. These observations are consistent with literature reports of SILM separations with different types of support membranes, with molecular simulations of RTIL characteristics near a solid interface and with reports of enhanced CO₂ solubility with confinement of non-ionic solvents. The presence of the pore surface appears to change the physical properties of these ionic liquids, perhaps through a reorganization of the cations and anions at the interface, increasing the free volume available for CO₂ absorption.

The results reported here raise a number of additional questions about this phenomenon. Do the property changes that occur with confinement depend upon the pore size and surface chemistry of the porous support and the structure of the ionic liquid? Are changes in free volume with confinement responsible for the observed changes in behavior? Do other gases exhibit similar increases in solubility and diffusivity? We plan to address these questions in our future work.

■ ASSOCIATED CONTENT

Supporting Information

Comparison of experimental and model pressures versus time values for one CO₂ uptake experiment (Figure S1) and summary of *H* and *D* values determined for CO₂ in bulk and

confined ionic liquids (Table S1). This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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